AI Playbook and Toolkit for Public Health Departments

Communicating Model Results and Uncertainty

ANL 350

Public health staff often need to communicate AI or model results to leaders, partners, the public, or operational teams. This module teaches learners how to explain model outputs, uncertainty, limitations, and appropriate use without overstating precision or creating false confidence. The module focuses on practical communication. Learners translate metrics, confidence, caveats, and validation findings into language that supports responsible decisions. The goal is to ensure that model results inform judgment rather than replace it.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling, communications, public-health-executive-leadership, epidemiology, governance-security

Learning Objectives

- Explain model results in plain language for nontechnical audiences.
- Distinguish uncertainty, variability, error, bias, and data limitations.
- Identify when model results should include confidence intervals, prediction intervals, caveats, or subgroup findings.
- Avoid overclaiming, causal overreach, and false precision in AI communications.
- Create a model communication brief for a public health decision-maker.

Module Overview

Public health staff often need to communicate AI or model results to leaders, partners, the public, or operational teams. This module teaches learners how to explain model outputs, uncertainty, limitations, and appropriate use without overstating precision or creating false confidence.

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Definitions

- Uncertainty: The degree to which an estimate, prediction, or conclusion may be imprecise or incomplete.
- Prediction interval: A range that estimates where a future observation may fall.
- Confidence interval: A range that expresses uncertainty around an estimated parameter.
- Calibration: How closely predicted probabilities match observed outcomes.
- Caveat: A clear statement of limitation, condition, or caution.

Why This Topic Matters for Public Health AI

Model outputs can influence urgent decisions. A forecast may affect staffing, a risk score may affect outreach, and a hotspot map may affect deployment of field teams. If uncertainty is not explained clearly, decision-makers may treat estimates as facts or ignore important limitations.

Public health communication must balance clarity and caution. Too much technical detail can obscure the message, but too little detail can hide uncertainty. A strong communication product explains what the model suggests, what it does not show, how reliable it is, and what human judgment or additional evidence is needed.

Public Health Example

An epidemiology team uses a time-series model to forecast emergency department visits for respiratory illness. The forecast shows a possible increase over the next two weeks, but the prediction interval is wide because reporting patterns changed after a holiday. A strong communication brief would state that visits may rise, explain the uncertainty, identify possible reasons for reduced confidence, and recommend preparedness actions that are proportional to the evidence.

The brief should not say that the model proves an outbreak is occurring. It should present the forecast as one input into surveillance judgment, alongside laboratory data, provider reports, syndromic trends, and field intelligence.

Technical and Operational Deep Dive

Model communication begins with the decision context. The same model output may be communicated differently to a technical team, health officer, program manager, community partner, or public audience. Technical users may need details about metrics and data limitations, while executives may need decision implications, confidence, and recommended actions.

Uncertainty should be explained visually and verbally. Trend lines, ranges, maps, and risk categories can help, but they can also imply precision that the model does not have. Labels should avoid unsupported certainty. Words such as likely, possible, estimated, preliminary, and subject to change can be helpful when used consistently.

Subgroup findings require special care. If model performance differs by age, race, ethnicity, language, geography, disability, housing status, or data source, communications should avoid presenting aggregate performance as universal. Decision-makers need to know whether a model is less reliable for some populations.

Risks, Failure Modes, and Guardrails

A major failure mode is false precision. A model may produce a number such as 82.4 percent risk, but the data and validation may not support that level of exactness. A guardrail is to use appropriate rounding, risk categories, and uncertainty statements.

Another failure mode is hiding limitations to make results look more actionable. This can erode trust if the model later performs poorly. A guardrail is to include a standard limitations section in model briefs and require review before results are shared outside the technical team.

Application to AI-Supported Workflows

Generative AI can help draft summaries of model findings, but the draft must be checked by analysts who understand the model. Predictive analytics, outbreak forecasts, risk stratification tools, and geospatial models all require communication practices that preserve uncertainty. RAG systems used for briefings should cite source reports and validation evidence.

Reflection Questions

Who needs to understand this model result?

What decision will the result support?

What uncertainty or limitation could change the decision?

Are there subgroup performance concerns?

What wording would prevent overclaiming?

Practical Exercise

Create a one-page model communication brief. Include the decision question, model result, uncertainty, validation status, limitations, subgroup considerations, recommended use, prohibited use, and next review date.

Expected Artifact or Evidence

Model communication brief

Plain-language uncertainty statement

Subgroup caveat if relevant

Review sign-off note

Expected Assignment

- Model communication brief
- Plain-language uncertainty statement
- Subgroup caveat if relevant
- Review sign-off note

Knowledge Check

- 1. What is the best reason to communicate uncertainty with model results?
- 2. What is the best reason to assign an accountability owner?
- 3. When should an AI-supported workflow be escalated for review?
- 4. Which practice best supports public accountability?
- 5. What should happen when data sources, policy, vendor configuration, or workflow use changes?

References and Resources

- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- NIST Artificial Intelligence Risk Management Framework and NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- WHO Ethics and Governance of Artificial Intelligence for Health guidance: https://www.who.int/publications/i/item/9789240029200
- OWASP Top 10 for Large Language Model Applications: https://owasp.org/www-project-top-10-for-large-language-model-applications/
- ONC health IT, interoperability, and information blocking resources: https://www.healthit.gov/
- Agency policies for privacy, public records, cybersecurity, procurement, civil rights, accessibility, language access, and emergency communications